

Post-Quantum TLS Measurement as Deployment Evidence

Abstract

Post-quantum TLS migration changes deployed protocol behavior across handshake size, certificate-chain composition, fallback paths, client support, and middlebox behavior. This manuscript frames migration evidence as a measurement object: traces, normalized observations, static reports, and scoped deployment evidence.

Methodology

The measurement pipeline separates endpoint capability, negotiated parameters, certificate-chain shape, fallback behavior, and report normalization. Each run records the probe configuration, endpoint response, cryptographic material, and failure label. The resulting tables keep raw observations close to the summary fields used for comparison.

Results

The current record gives a deployment-oriented grammar for PQ-TLS readiness: what was offered, what was negotiated, how the certificate path changed, and where interoperability failed. The code repository contains the public observability framework.